

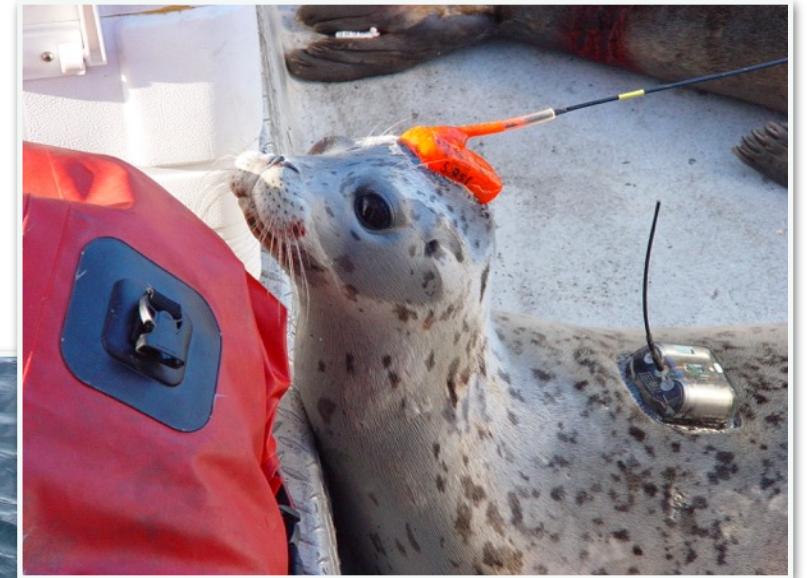
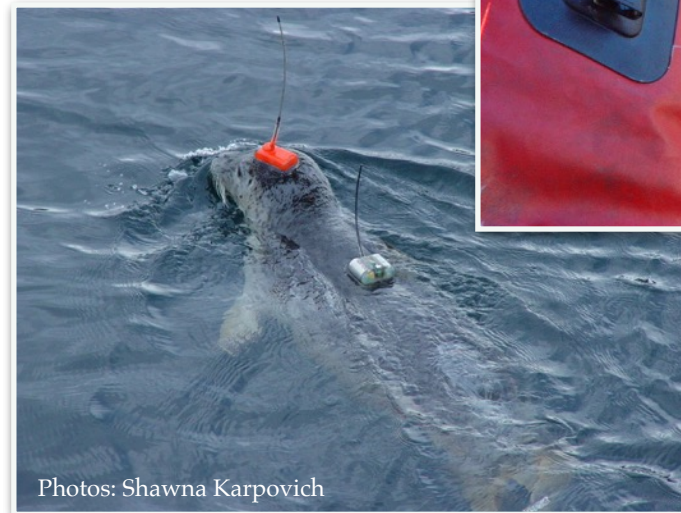
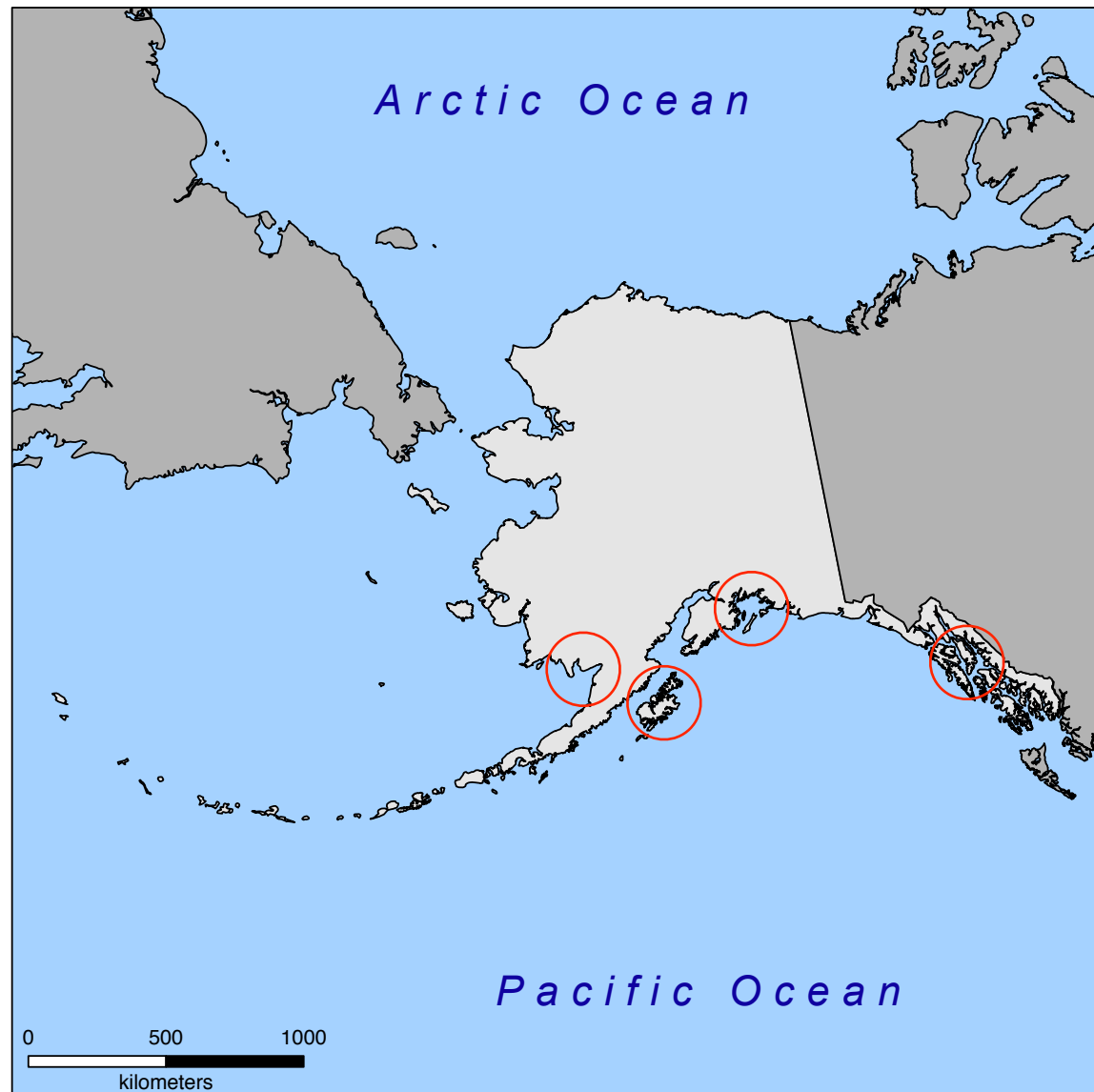
STATISTICAL MODELS FOR ANIMAL TELEMETRY DATA WITH APPLICATIONS TO HARBOR SEALS IN ALASKA

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Marine Mammal Laboratory Noontime Seminar

02 February 2017

Existing data

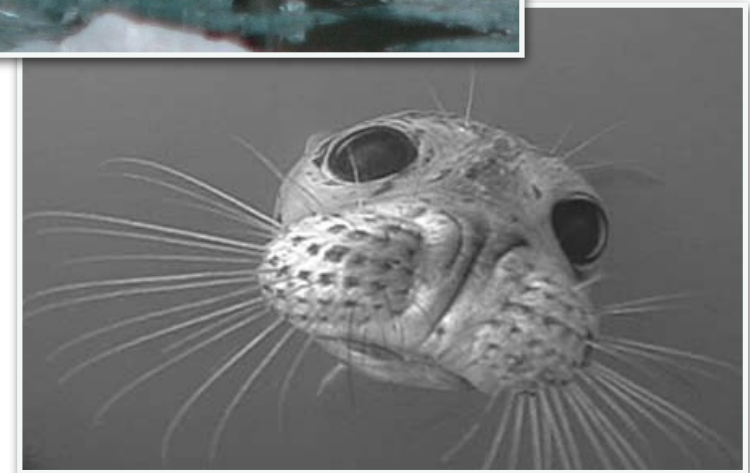


Preview

- Research objectives
 - Aquatic resource selection and space use
 - Terrestrial space use and temporal haul-out behavior
- Modeling goals
 - Use all available data
 - Account for uncertainty
 - Autocorrelation
 - Movement constraints



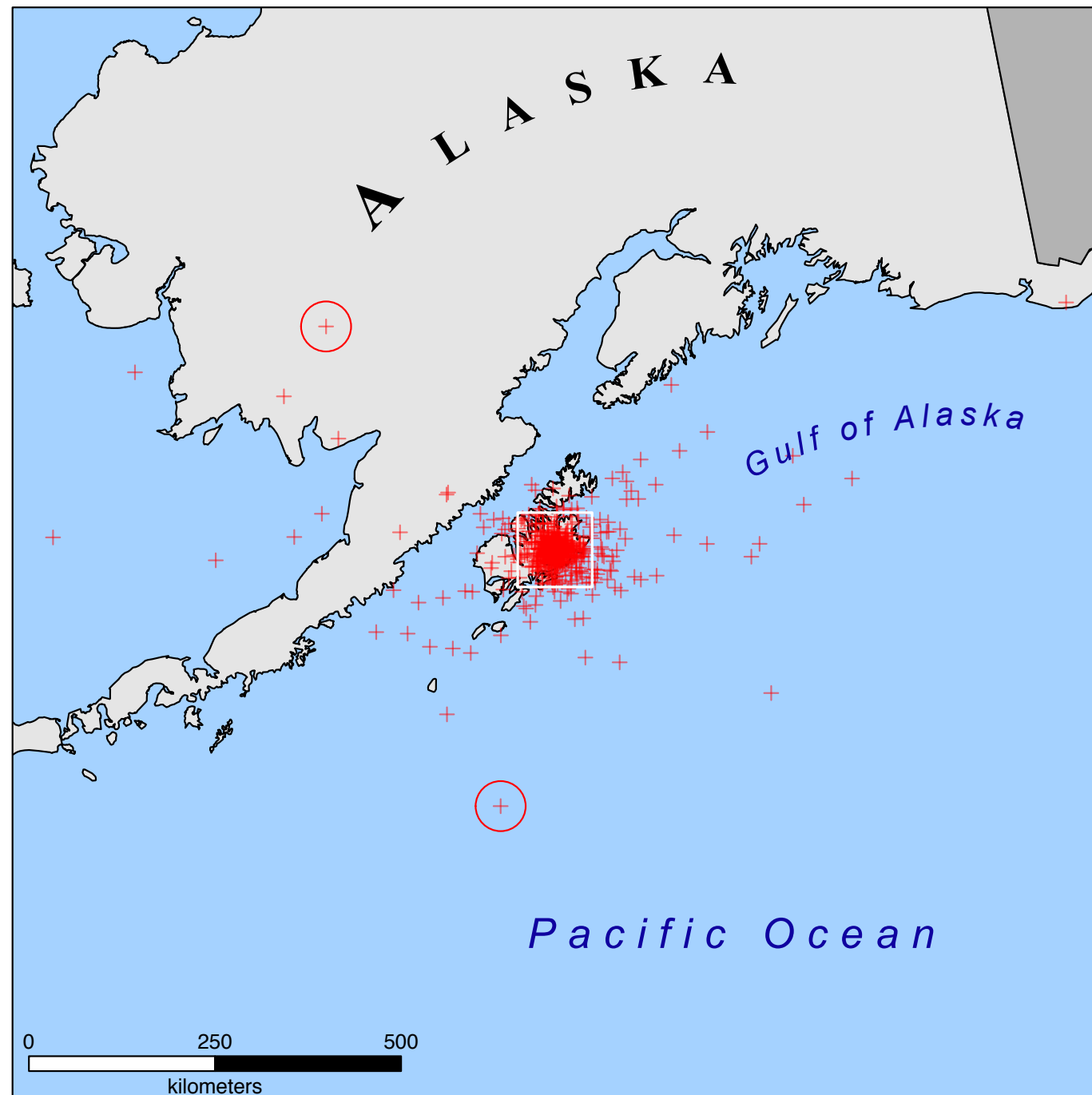
Photo: Shawna Karpovich



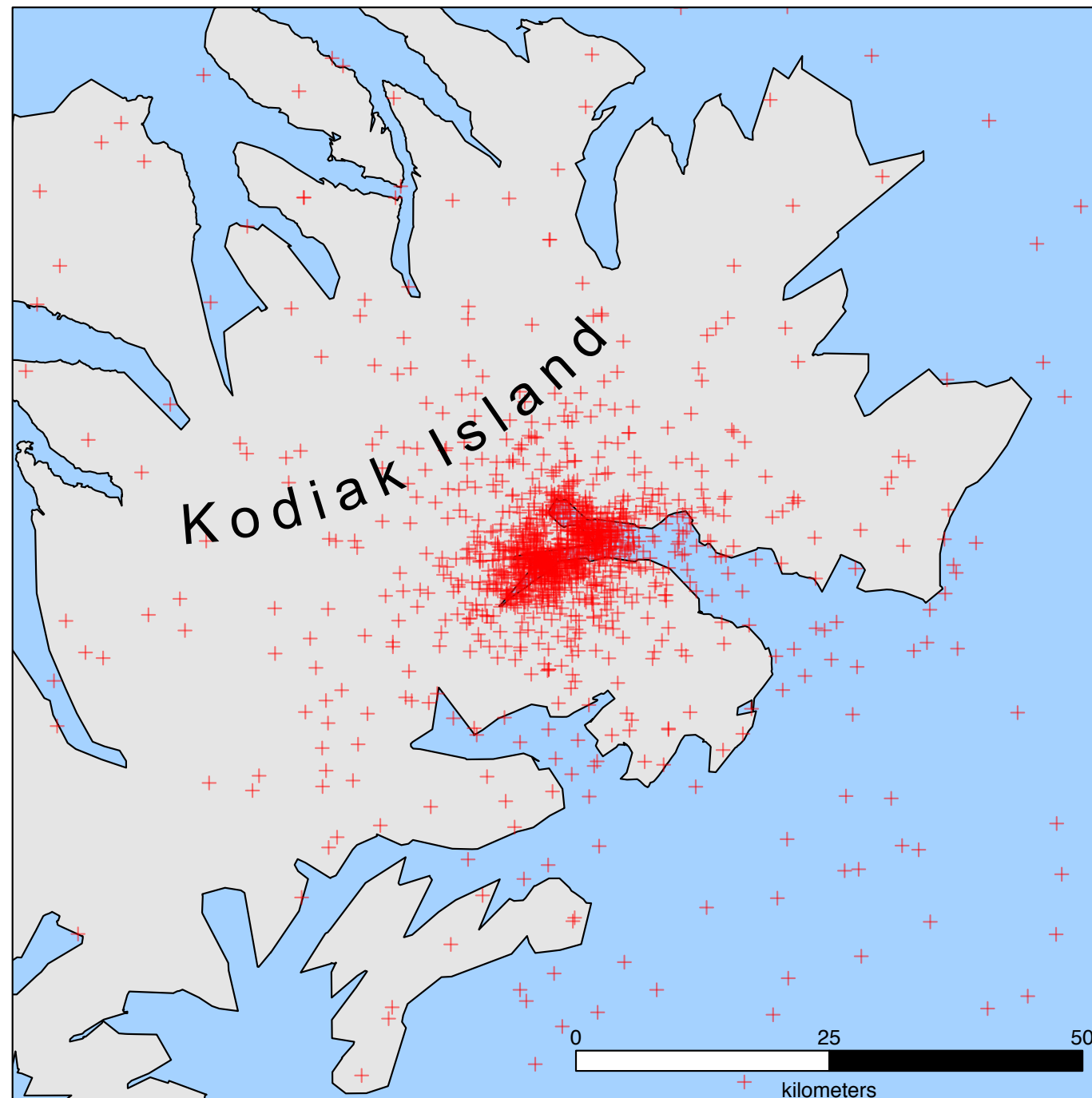
Aquatic resource selection and space use

Terrestrial space use and temporal haul-out behavior

Harbor seal telemetry locations



Harbor seal telemetry locations

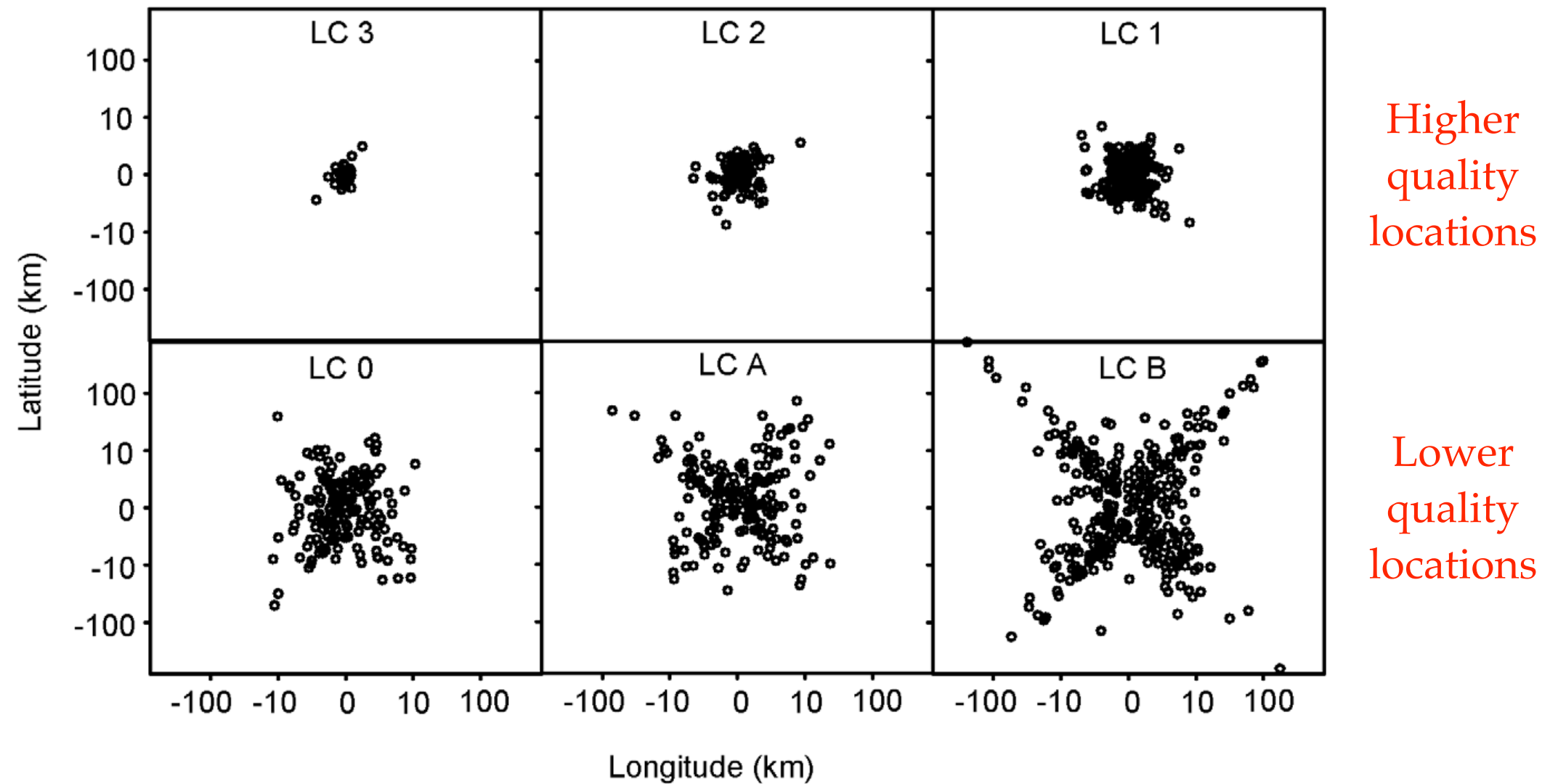


Simple fixes?

- “Snap” locations to water
- Speed filter
- Eliminate locations on land



Argos telemetry error



Hierarchical resource selection model

Observation model

$$\mathbf{s}(t) \sim \begin{cases} t(\boldsymbol{\mu}(t), \boldsymbol{\Sigma}, \nu), & \text{with prob. } p(t) \\ t(\boldsymbol{\mu}(t), \tilde{\boldsymbol{\Sigma}}, \nu), & \text{with prob. } 1 - p(t) \end{cases}$$

Process model

$$\boldsymbol{\mu}(t) \sim \frac{\exp \{ \mathbf{x}'(\boldsymbol{\mu}(t)) \boldsymbol{\beta} - \eta(\boldsymbol{\mu}(t), \boldsymbol{\mu}(t - \Delta t)) \}}{\int_{\mathcal{M}} \exp \{ \mathbf{x}'(\boldsymbol{\mu}) \boldsymbol{\beta} - \eta(\boldsymbol{\mu}, \boldsymbol{\mu}(t - \Delta t)) \} d\boldsymbol{\mu}}$$

$\mathbf{s}(t)$ — telemetry location at time t

$\boldsymbol{\mu}(t)$ — true location at time t

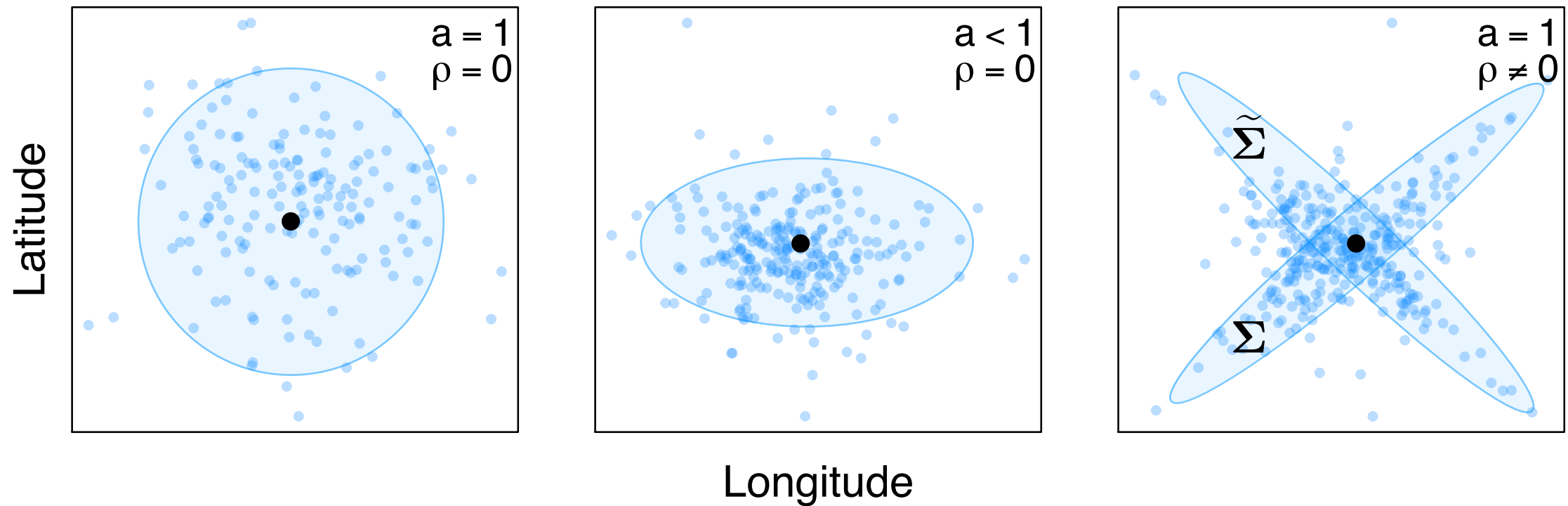
$\mathcal{M}$ — support of movement process

Observation model

Scale matrices

$$\Sigma = \sigma^2 \begin{bmatrix} 1 & \rho\sqrt{a} \\ \rho\sqrt{a} & a \end{bmatrix}, \quad \tilde{\Sigma} = \sigma^2 \begin{bmatrix} 1 & -\rho\sqrt{a} \\ -\rho\sqrt{a} & a \end{bmatrix}$$

Possible error patterns:



Process model

$$\boldsymbol{\mu}(t) \sim \frac{\exp \{ \mathbf{x}'(\boldsymbol{\mu}(t)) \boldsymbol{\beta} - \eta(\boldsymbol{\mu}(t), \boldsymbol{\mu}(t - \Delta t)) \}}{\int_{\mathcal{M}} \exp \{ \mathbf{x}'(\boldsymbol{\mu}) \boldsymbol{\beta} - \eta(\boldsymbol{\mu}, \boldsymbol{\mu}(t - \Delta t)) \} d\boldsymbol{\mu}}$$

Movement model

$$\eta(\boldsymbol{\mu}(t), \boldsymbol{\mu}(t - \Delta t)) = \frac{d(\boldsymbol{\mu}(t), \boldsymbol{\mu}(t - \Delta t))}{\Delta t \phi}$$

$\mathbf{x}(\boldsymbol{\mu}(t))$ — resource covariates at location $\boldsymbol{\mu}(t)$

$\boldsymbol{\beta}$ — resource selection coefficients

$\eta(\cdot)$ — movement model

$d(\boldsymbol{\mu}(t), \boldsymbol{\mu}(t - \Delta t))$ — distance between $\boldsymbol{\mu}(t)$ and $\boldsymbol{\mu}(t - \Delta t)$

Δt — time elapsed between $\boldsymbol{\mu}(t)$ and $\boldsymbol{\mu}(t - \Delta t)$

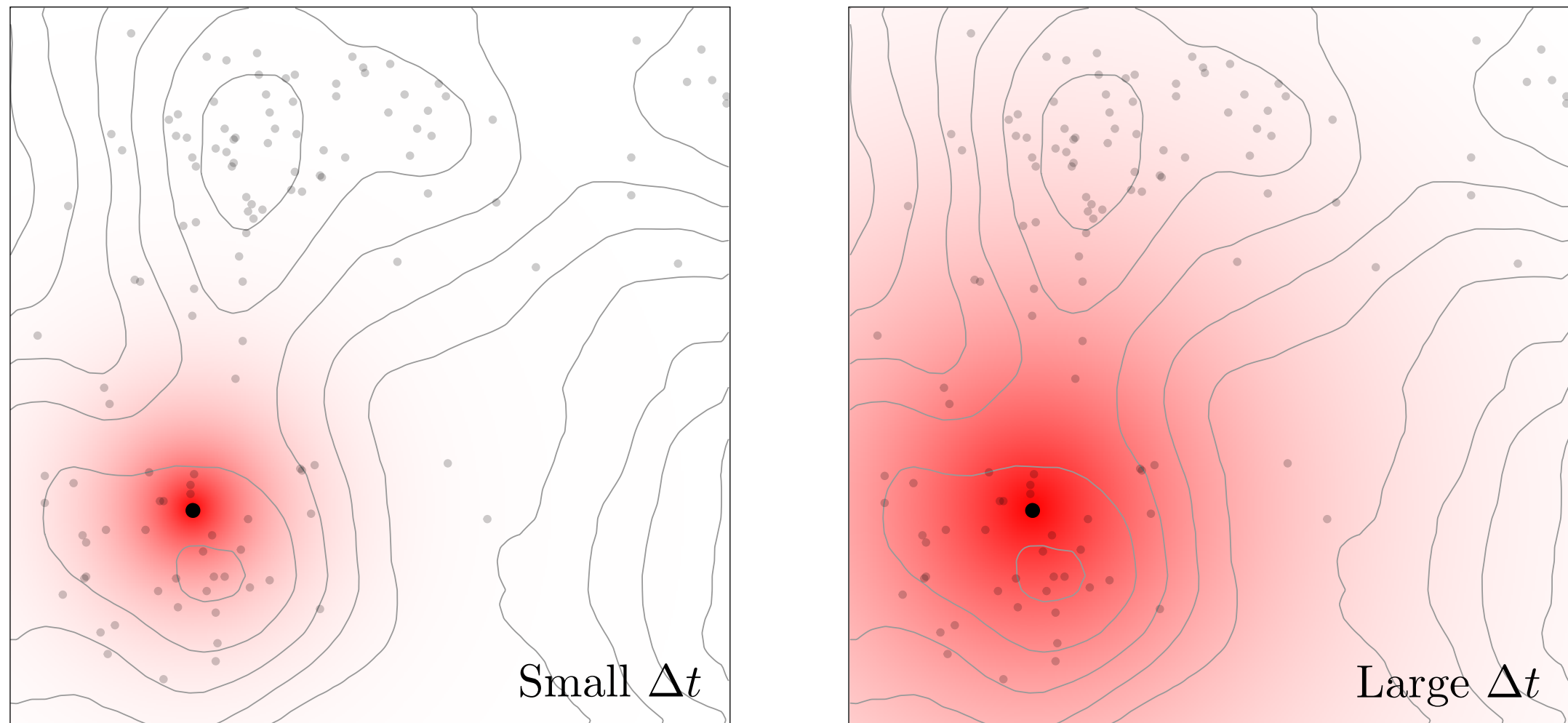
ϕ — movement rate

Johnson et al. 2008. A general framework for the analysis of animal resource selection from telemetry data. *Biometrics* 64:968–976.

Forester et al. 2009. Accounting for animal movement in estimation of resource selection functions: sampling and data analysis.

Ecology 90:3554–3565.

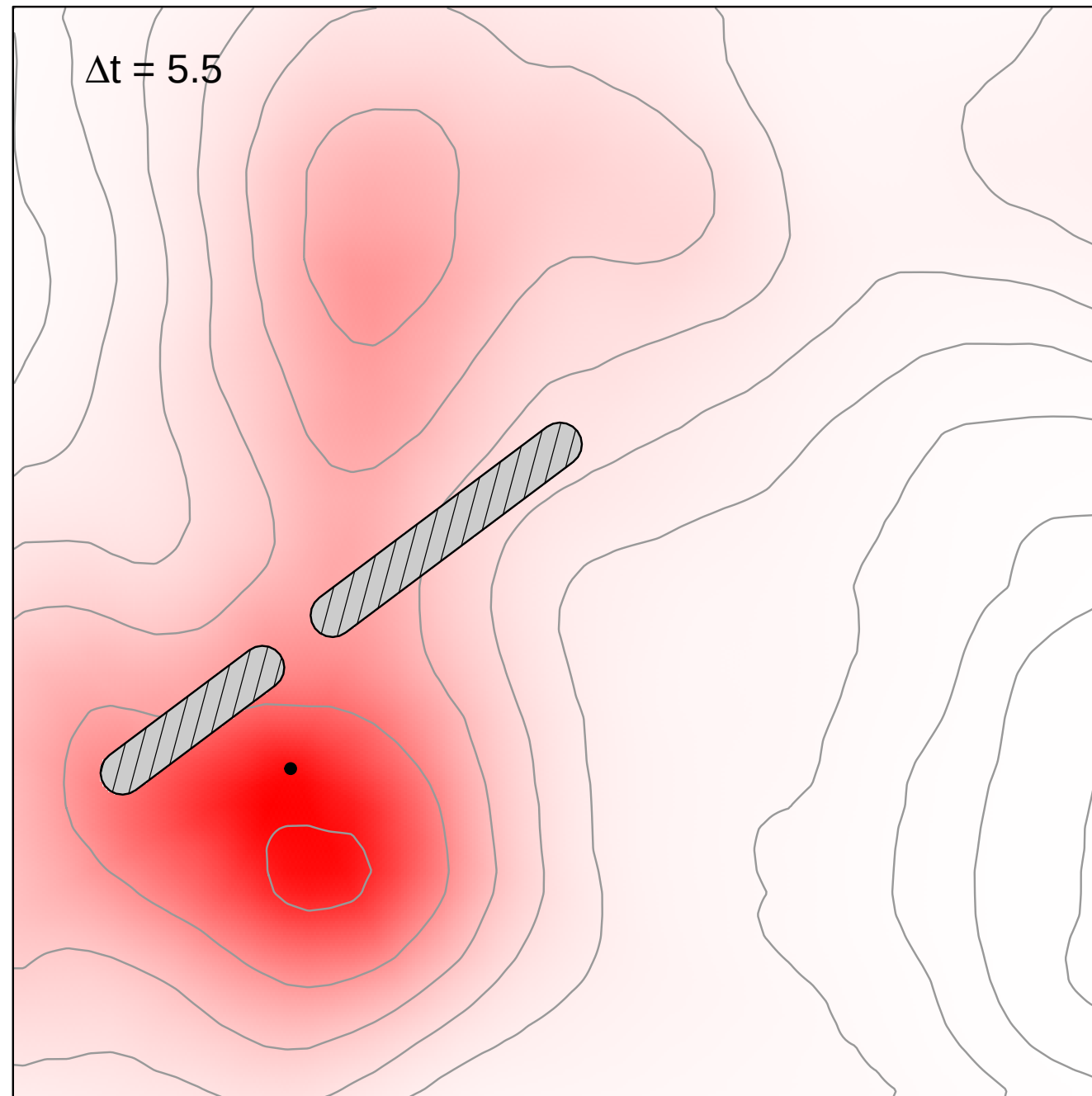
Movement and resource availability



Movement model:

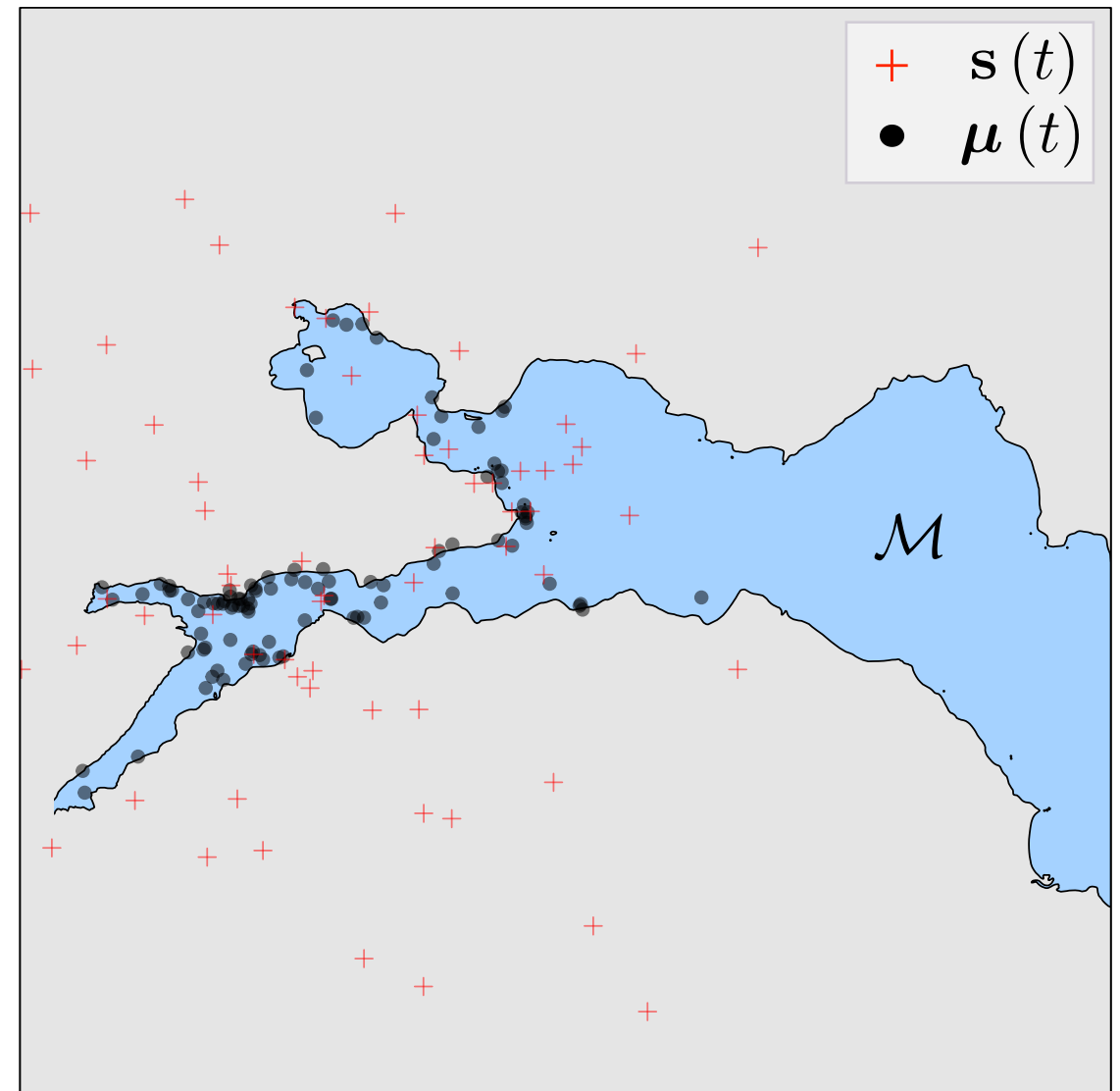
- Defines resource availability
- Accounts for autocorrelation

Probability of use



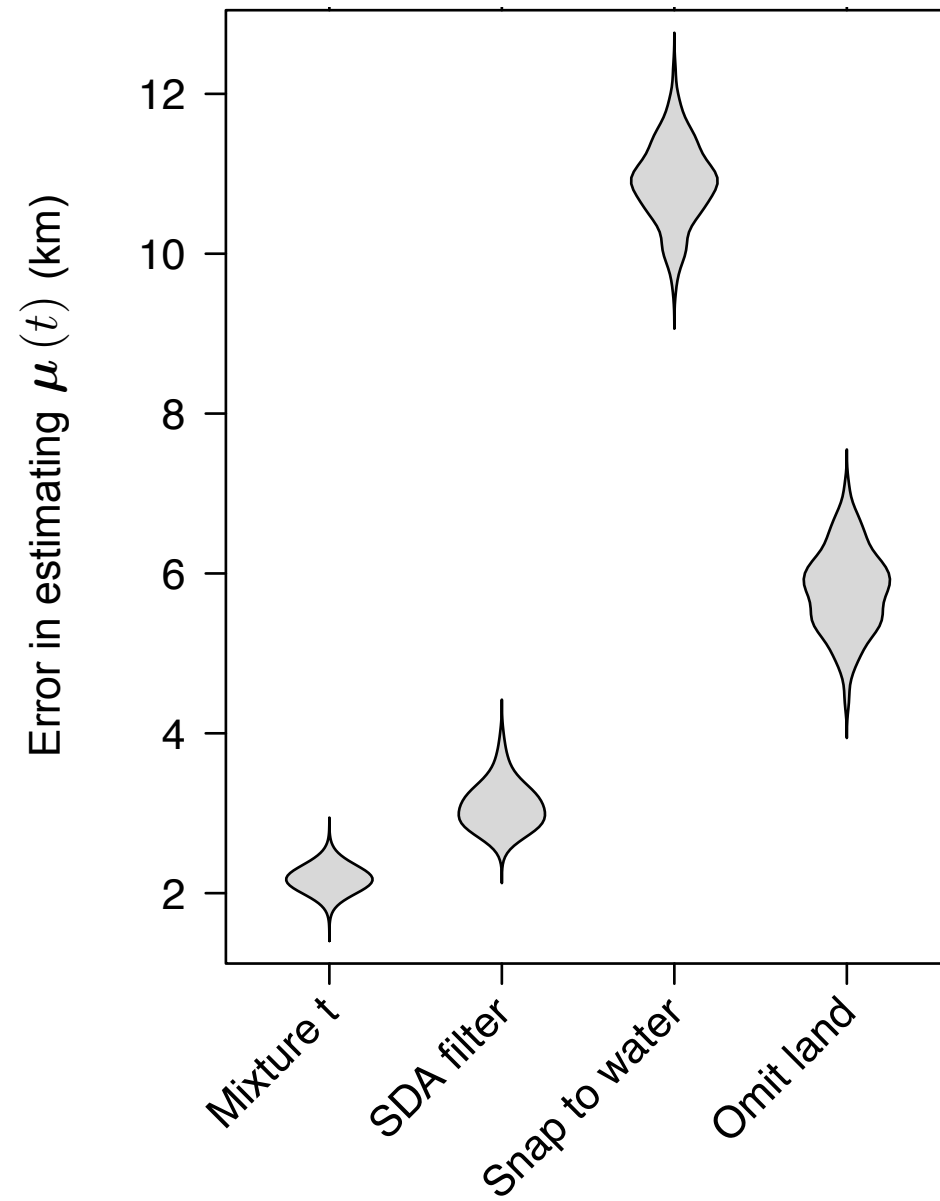
Error estimation

- Models typically require prior knowledge about error
- Discrepancy in support of $s(t)$ and $\mu(t)$ allow estimation of telemetry error

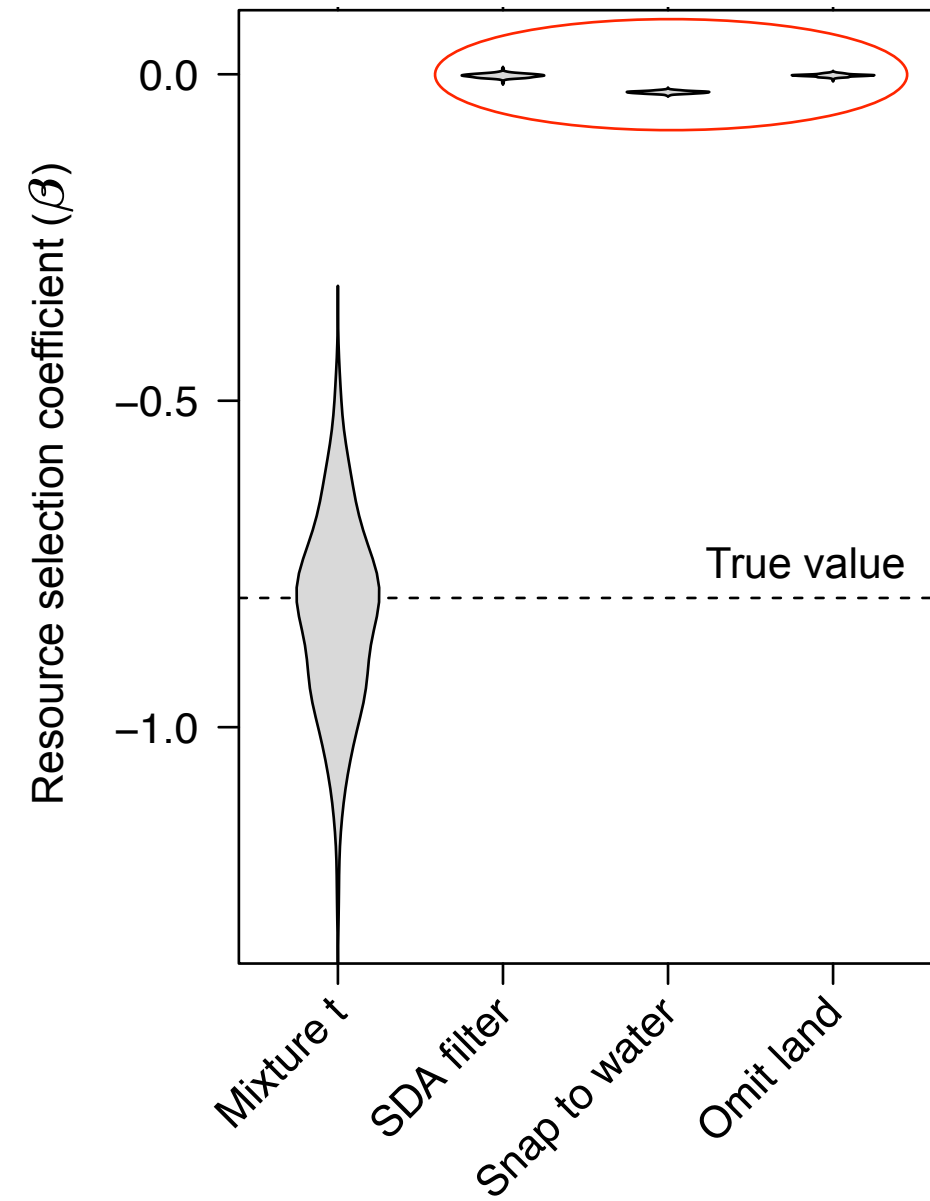


Simulation study

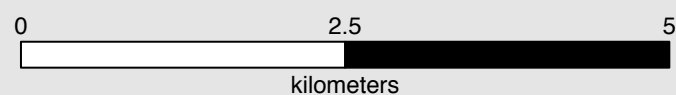
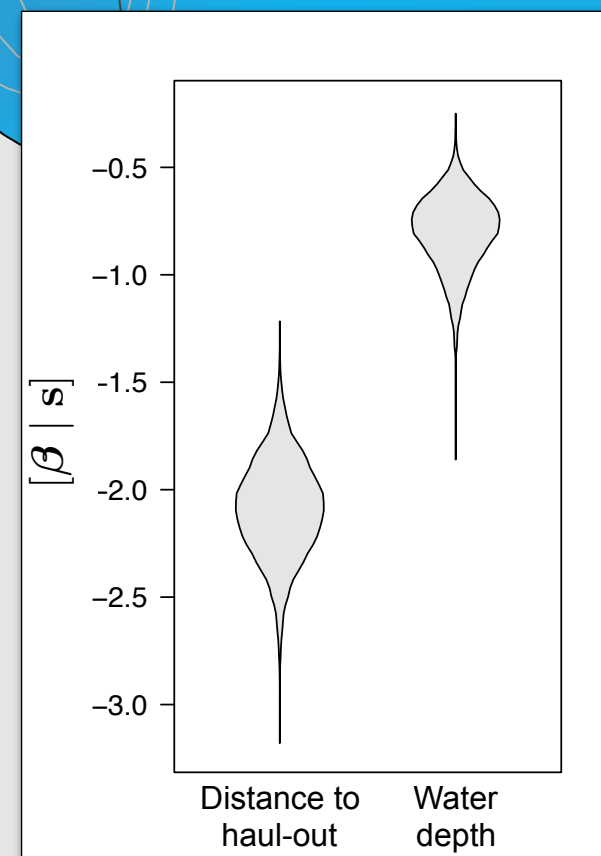
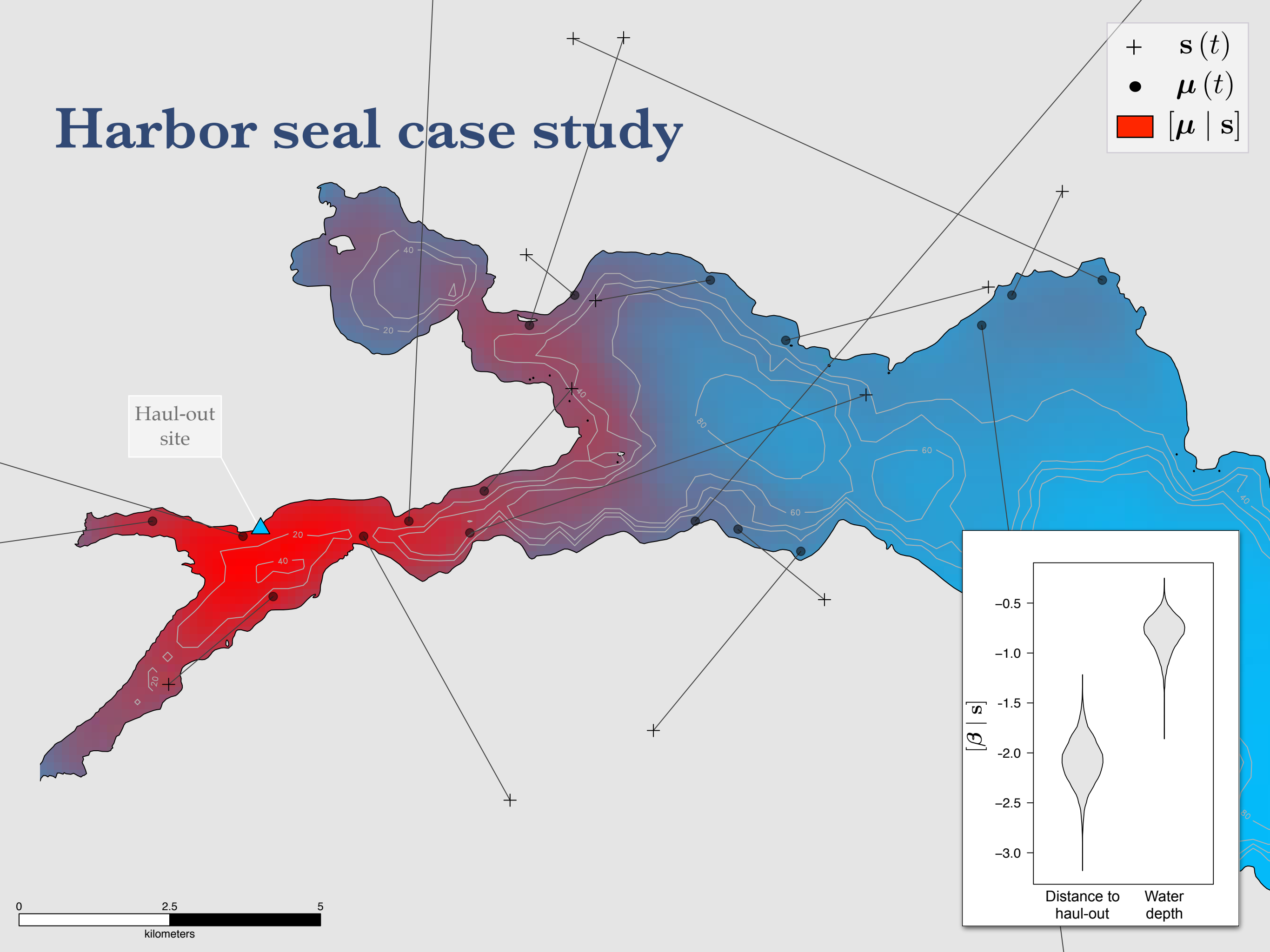
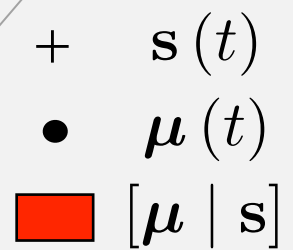
Estimation of $\mu(t)$



Estimation of β



Harbor seal case study



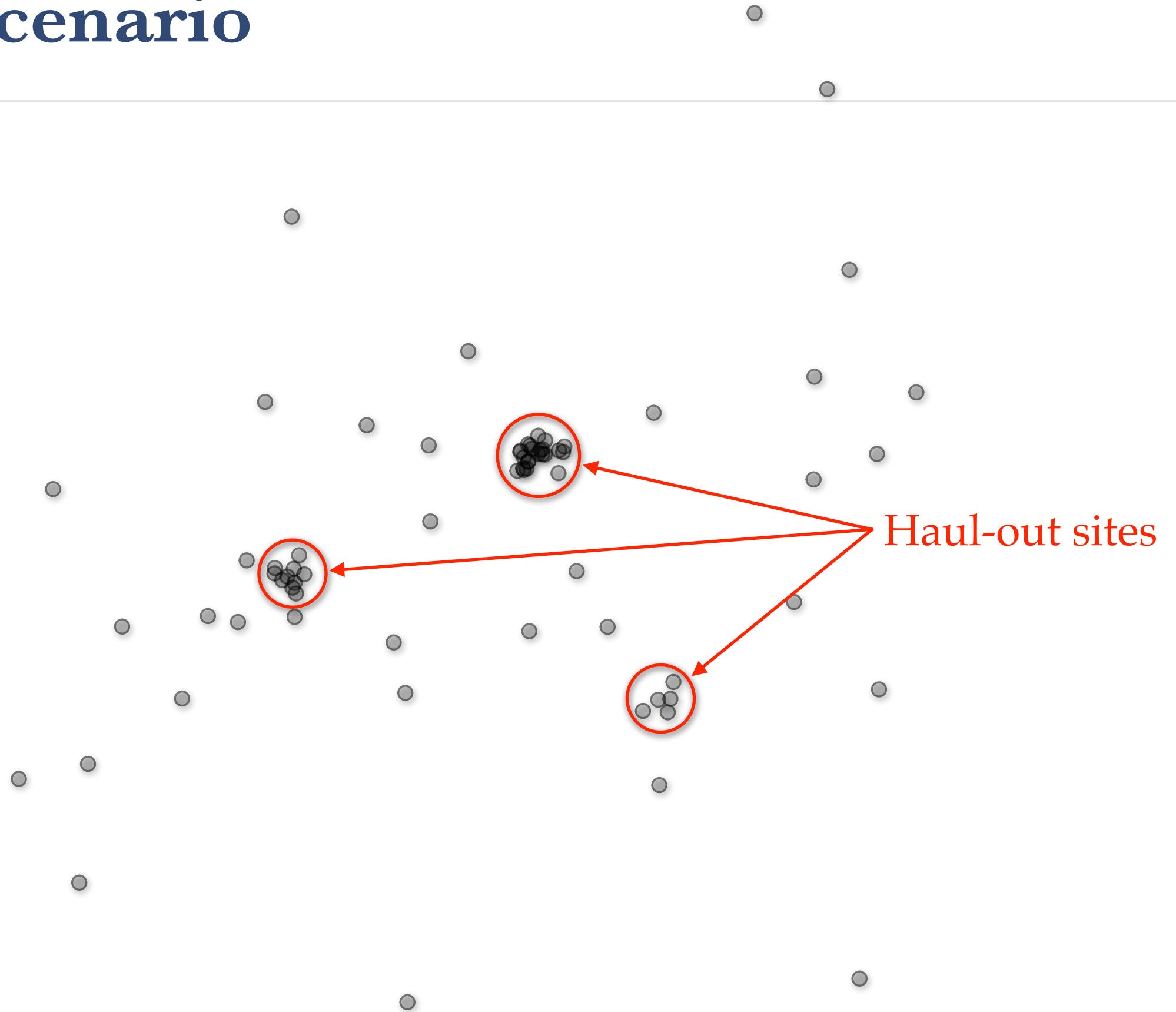
Conclusions

- Can estimate resource selection despite massive error
 - Account for telemetry location error
 - Animal movement
- Ignoring details leads to biased, over-confident inference
- Constraints are challenging but beneficial

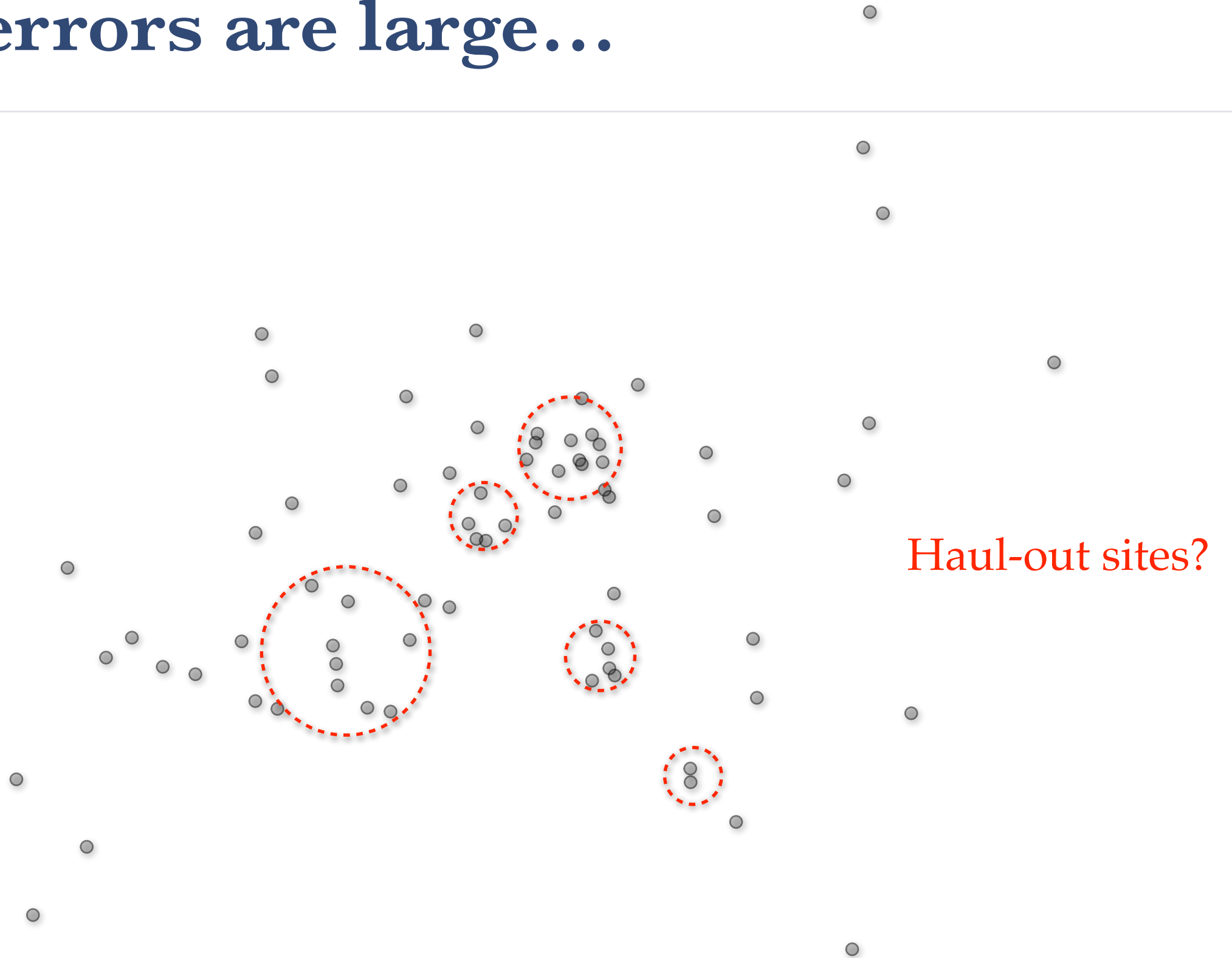
Aquatic resource selection and space use

Terrestrial space use and temporal haul-out behavior

Ideal scenario



When errors are large...

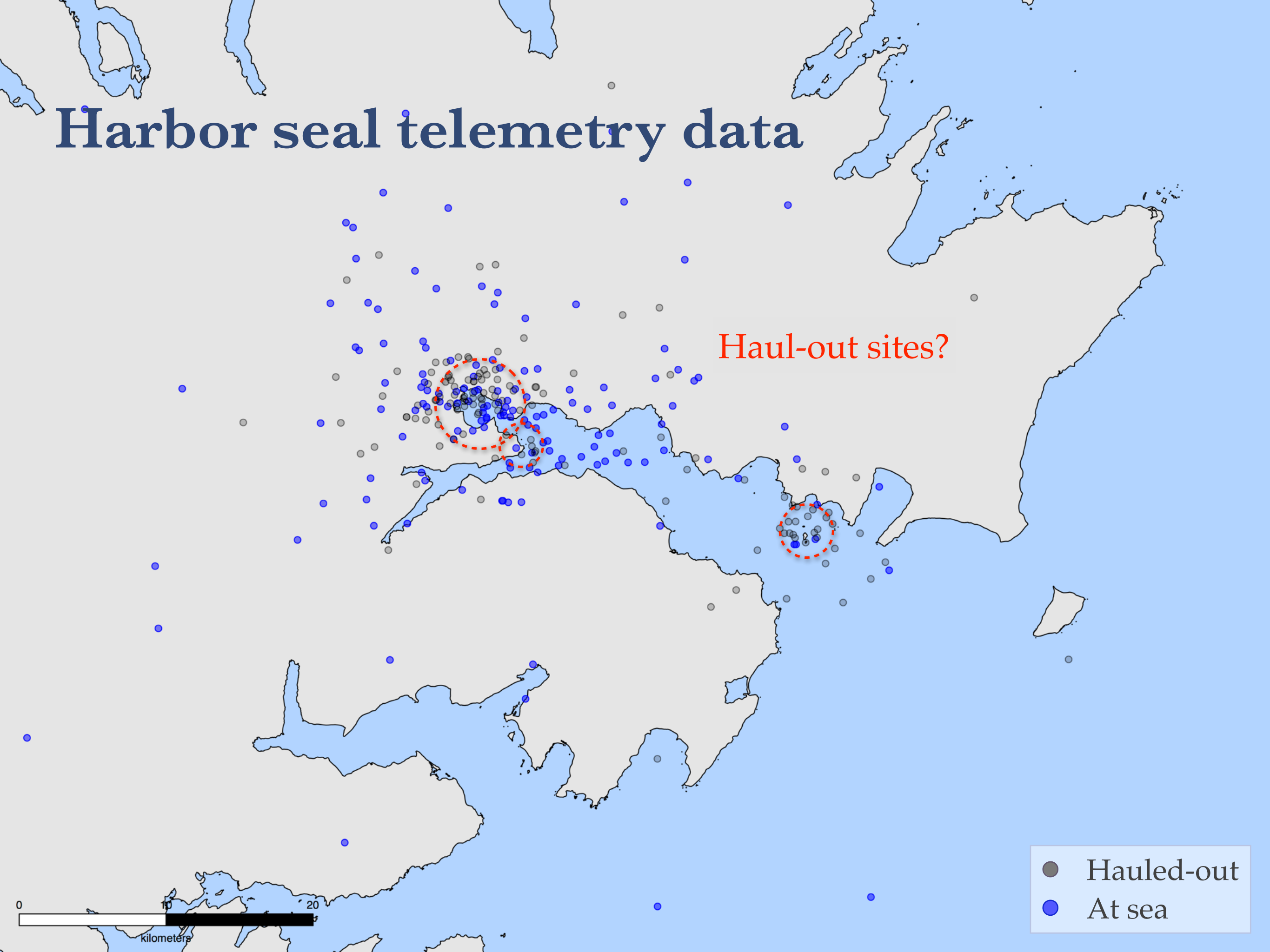


Harbor seal telemetry data

Haul-out sites?

- Hauled-out
- At sea

0 10 20
kilometers



Observation model

Hauled-out

$$\mathbf{s}(t) \sim \begin{cases} \mathcal{N}(\tilde{\boldsymbol{\mu}}(t), \boldsymbol{\Sigma}), & \text{with prob. } p(t) \\ \mathcal{N}(\tilde{\boldsymbol{\mu}}(t), \tilde{\boldsymbol{\Sigma}}), & \text{with prob. } 1 - p(t) \end{cases}$$

At sea

$$\mathbf{s}(t) \sim \begin{cases} \mathcal{N}(\tilde{\boldsymbol{\mu}}(t), \sigma_{\mu}^2 \mathbf{I} + \boldsymbol{\Sigma}), & \text{with prob. } p(t) \\ \mathcal{N}(\tilde{\boldsymbol{\mu}}(t), \sigma_{\mu}^2 \mathbf{I} + \tilde{\boldsymbol{\Sigma}}), & \text{with prob. } 1 - p(t) \end{cases}$$

$\mathbf{s}(t)$ — telemetry location at time t

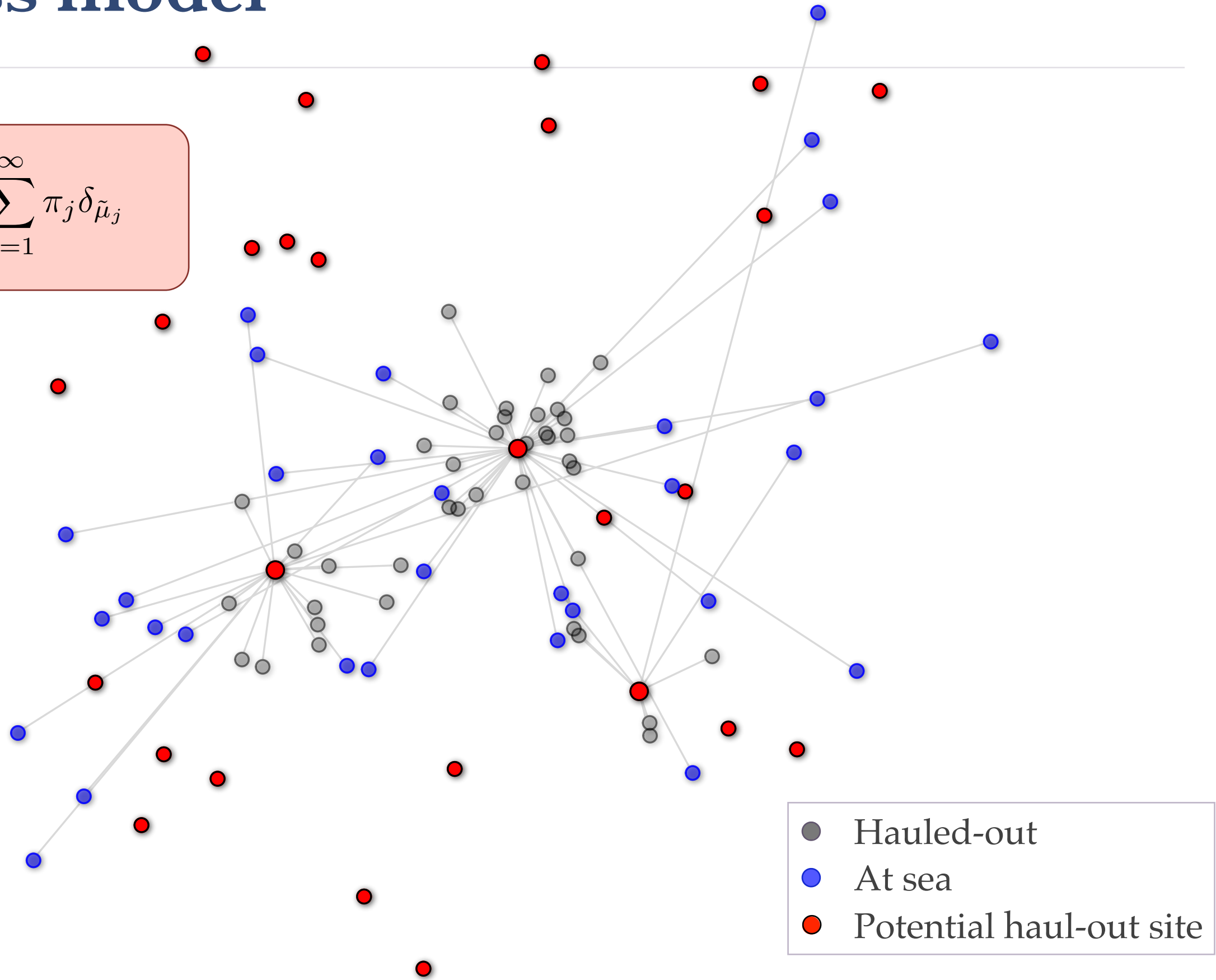
$\tilde{\boldsymbol{\mu}}(t)$ — haul-out location at time t

$\boldsymbol{\Sigma}$ and $\tilde{\boldsymbol{\Sigma}}$ — telemetry error covariance matrix

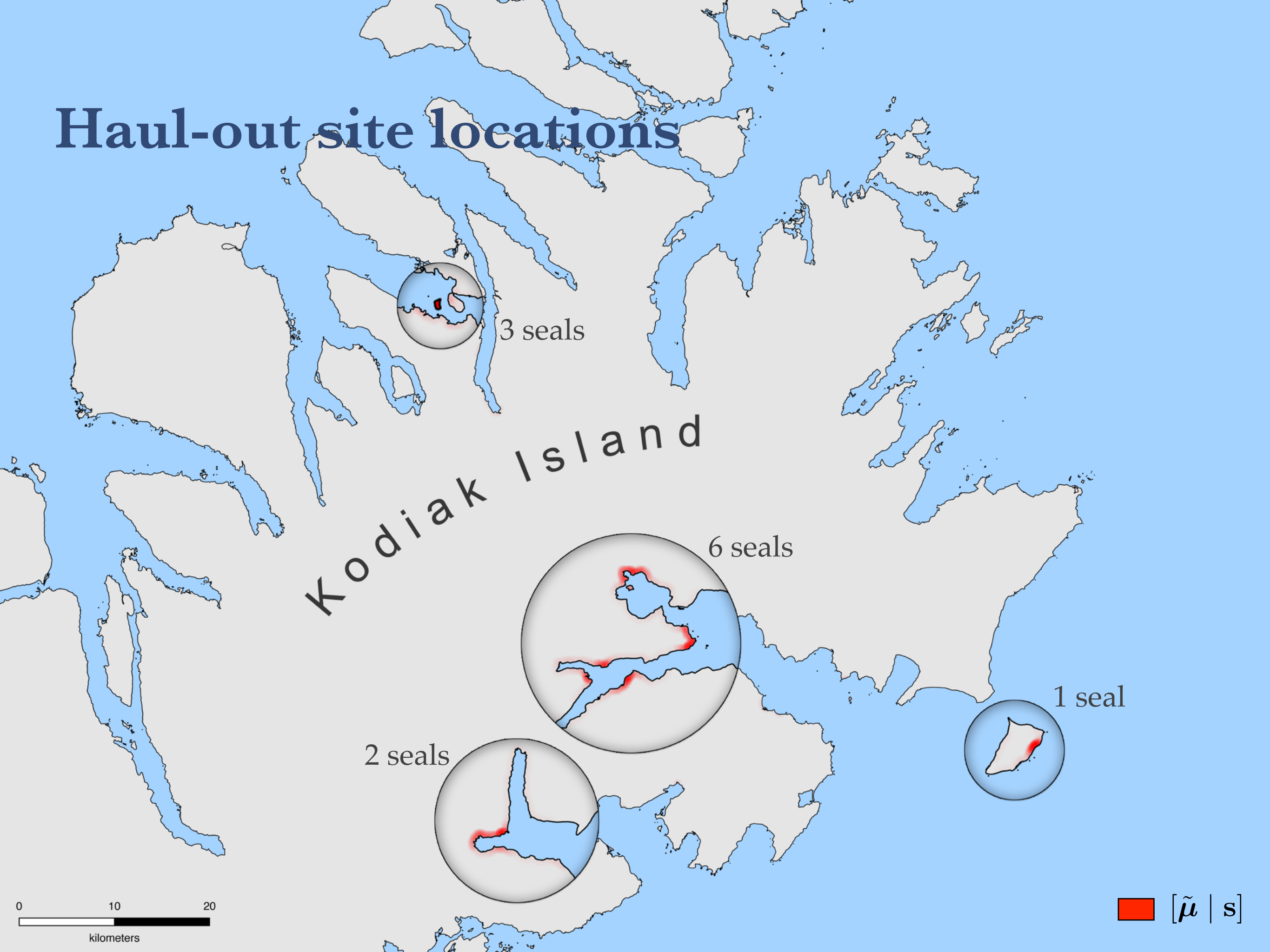
σ_{μ}^2 — movement parameter

Process model

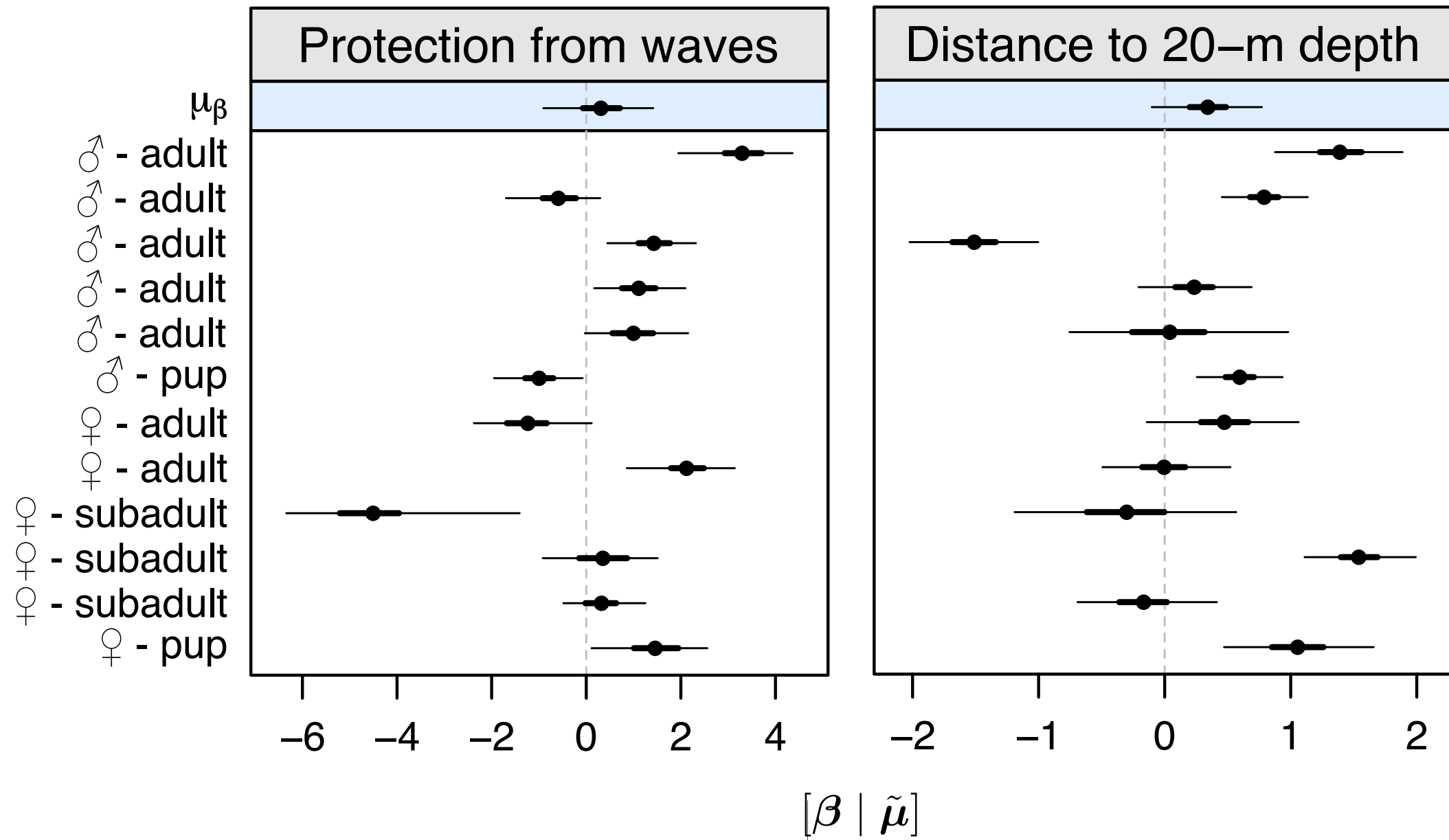
$$\tilde{\mu}(t) \sim \sum_{j=1}^{\infty} \pi_j \delta_{\tilde{\mu}_j}$$



Haul-out site locations



Haul-out site selection



Brost et al. *In review*. Model-based clustering reveals patterns in central place use of a marine top predator. *Journal of Animal Ecology*.

Temporal haul-out behavior

Observation model

$$y_i(t) \sim \text{Bernoulli}(p_i(t))$$

Process model

$$\begin{aligned} p_i(t) &= \Phi(\mathbf{x}_i(t)' \boldsymbol{\beta}_i + \mathbf{w}_i(t)' \boldsymbol{\alpha}_i) \\ \boldsymbol{\beta}_i &\sim \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta) \end{aligned}$$

$y_i(t)$ — wet/dry status at time t

$p_i(t)$ — probability $y_i(t) = 1$

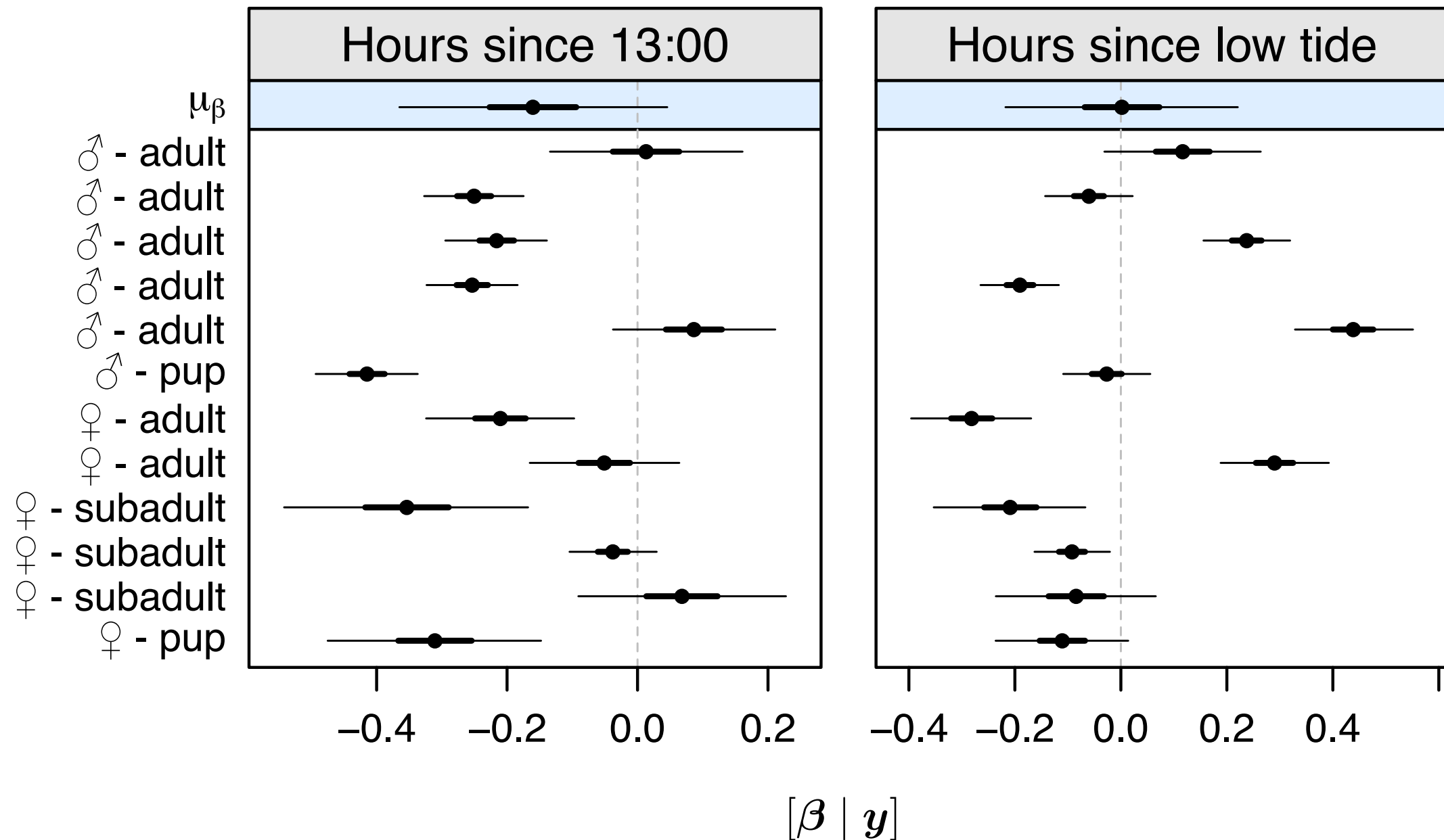
$\mathbf{x}_i(t)$ — environmental conditions at time t

$\boldsymbol{\beta}_i$ — coefficients for individual i

$\mathbf{w}_i(t)$ — basis functions evaluated at time t

$\boldsymbol{\alpha}_i$ — basis coefficients

Temporal haul-out behavior

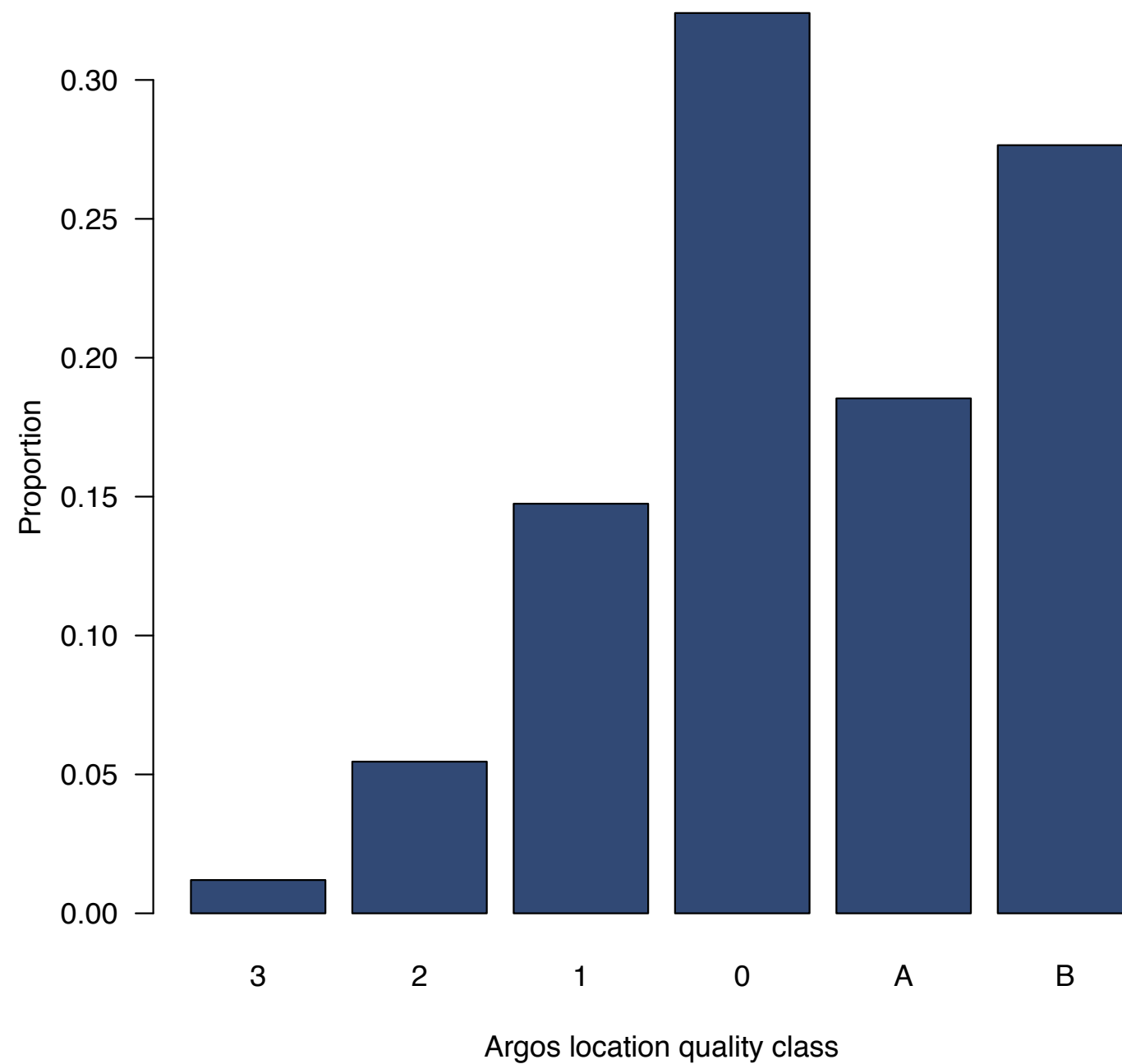


Conclusions

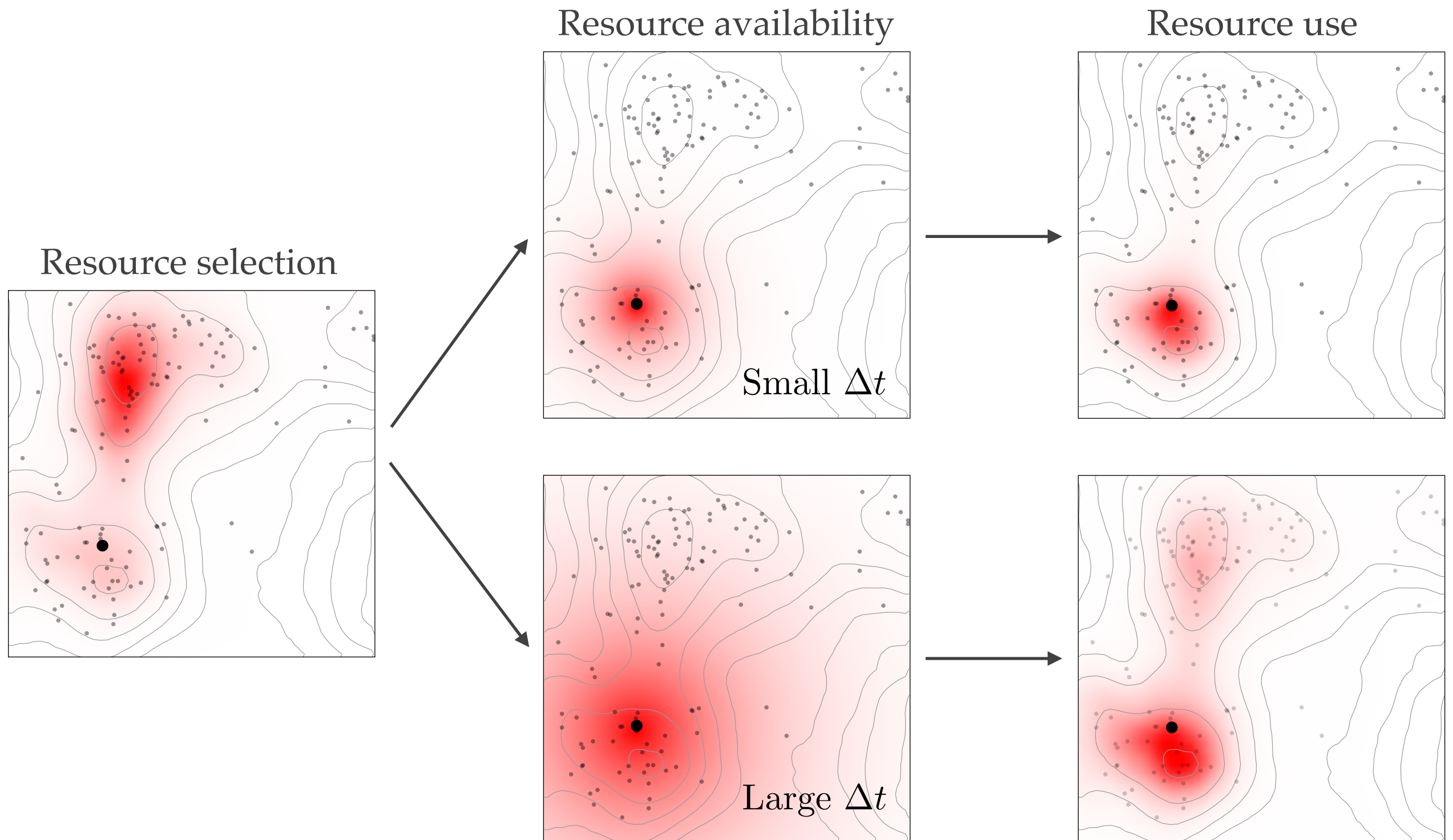
- Account for multiple sources of uncertainty
- Estimate location of haul-out sites outside of aerial surveys

QUESTIONS?

Argos telemetry error



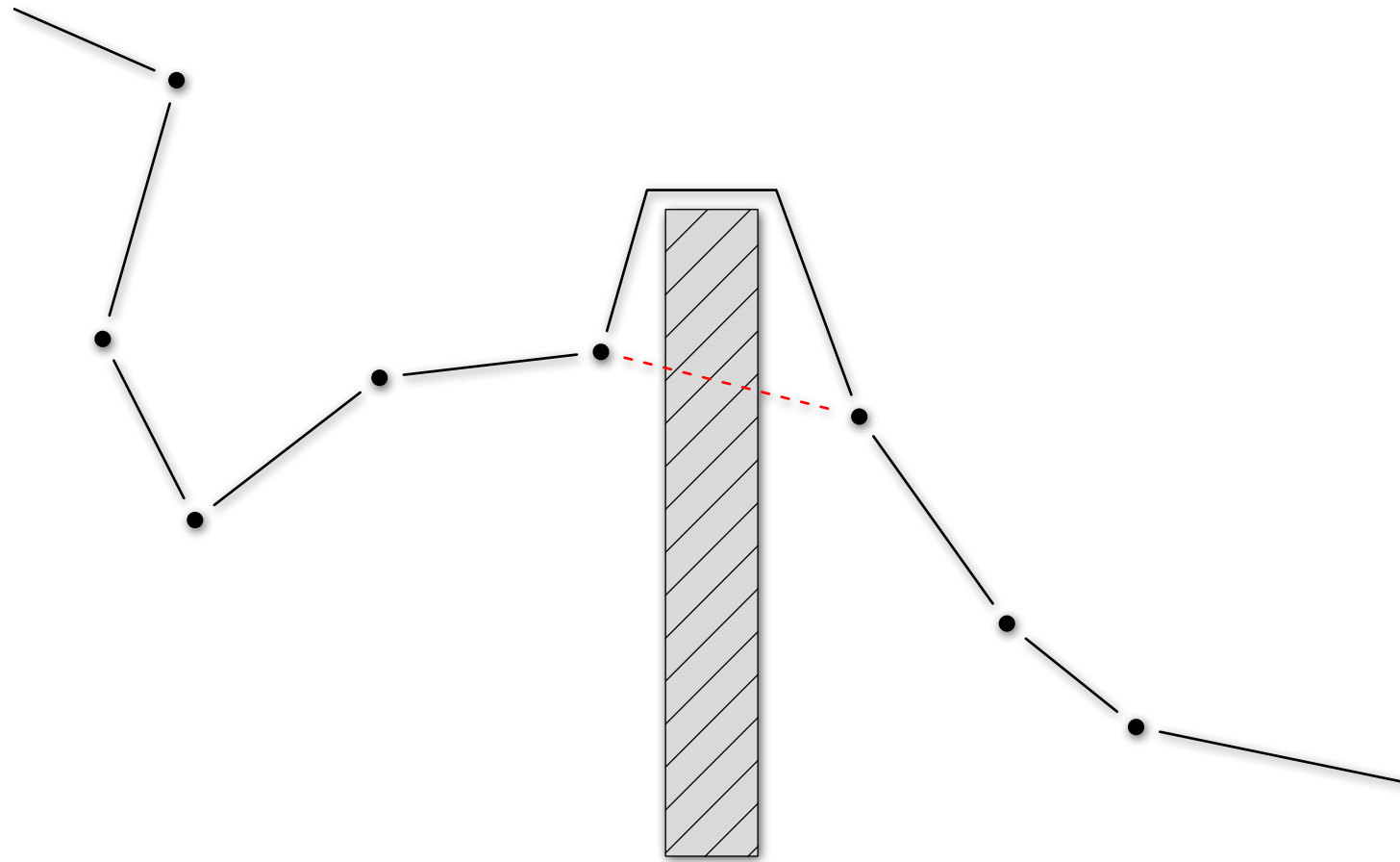
Process model



Movement constraints

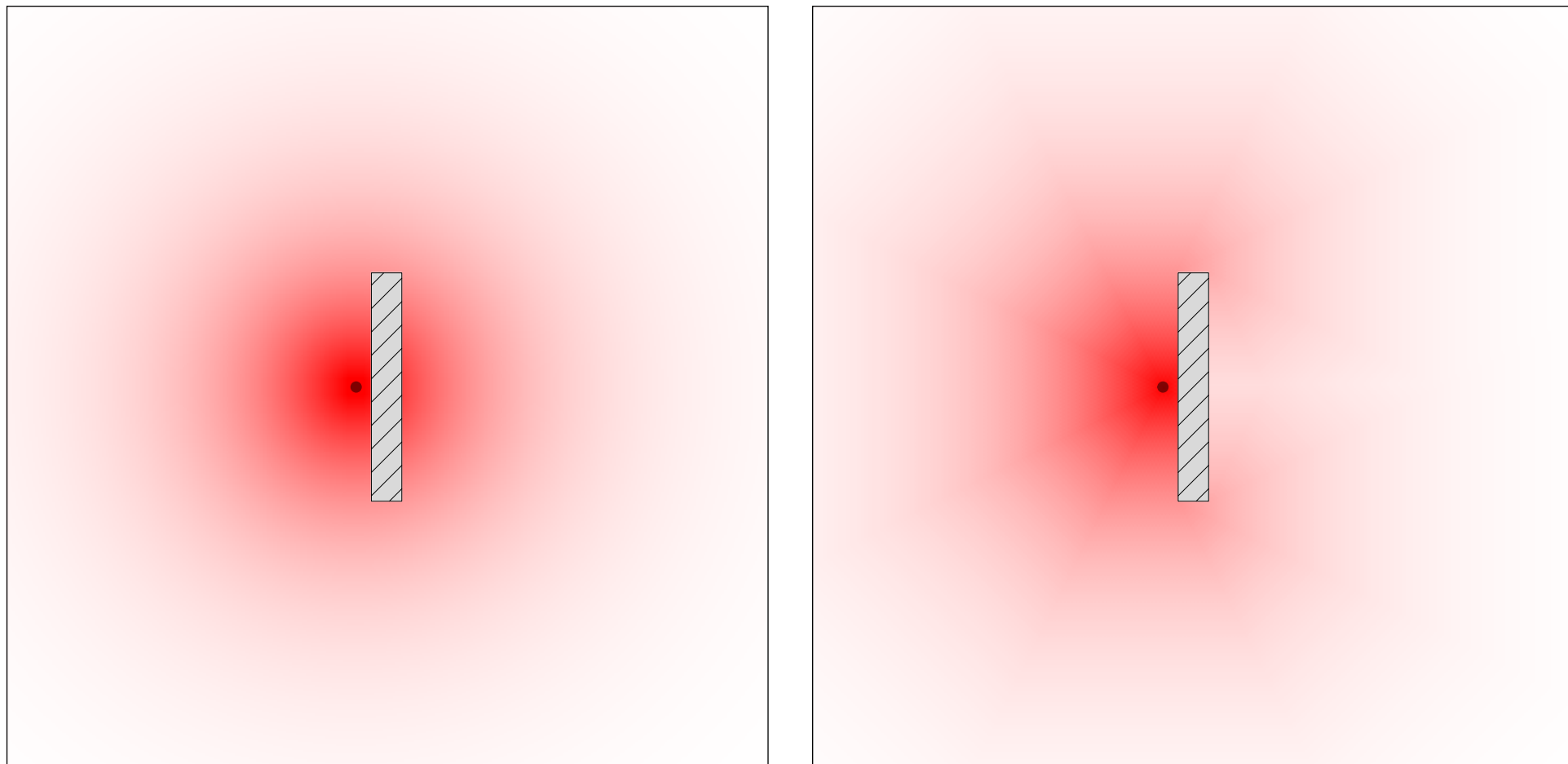
Movement model

$$\eta(\boldsymbol{\mu}(t), \boldsymbol{\mu}(t - \Delta t)) = \frac{d(\boldsymbol{\mu}(t), \boldsymbol{\mu}(t - \Delta t))}{\Delta t \phi}$$

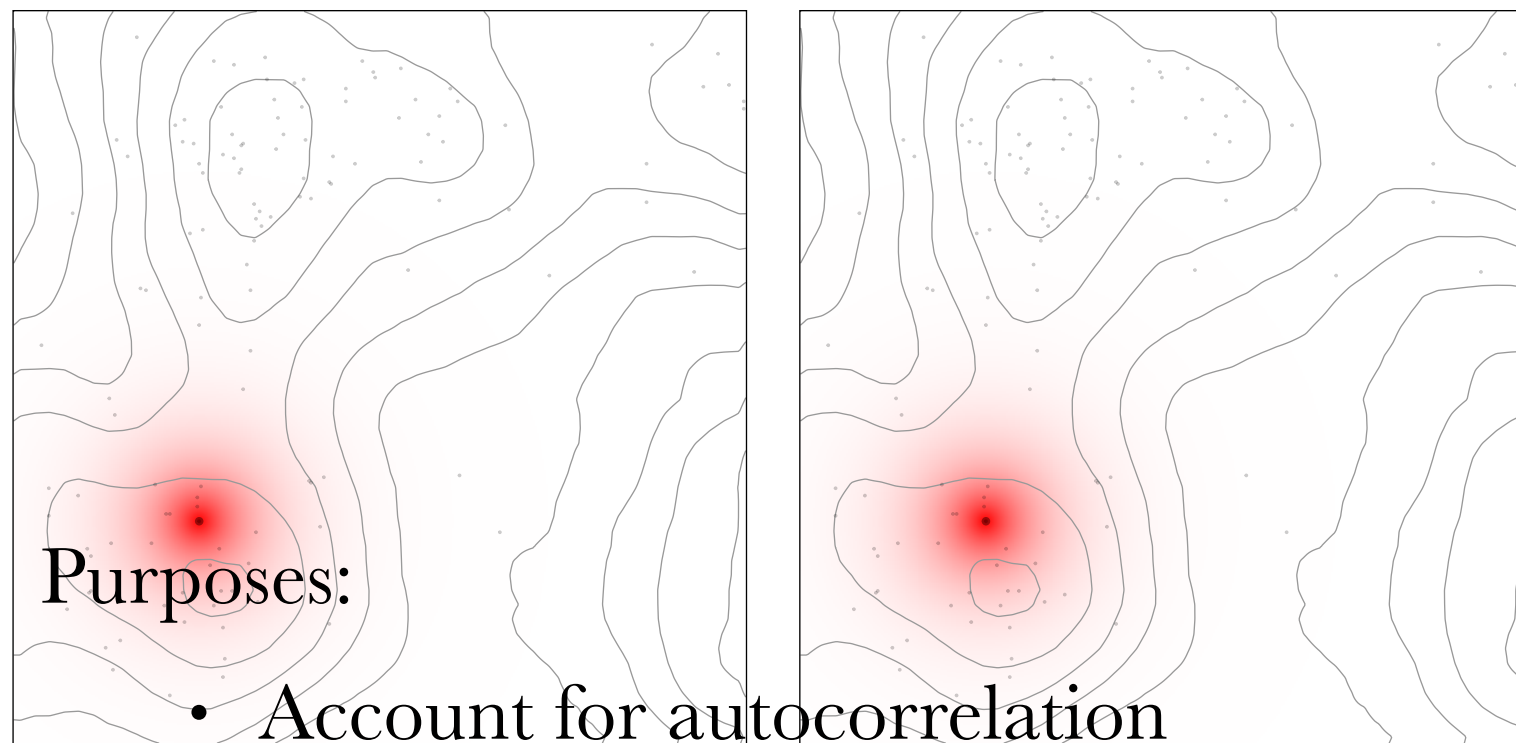


Process model

$$\mu(t) \sim \frac{\exp \{ \mathbf{x}'(\mu(t)) \boldsymbol{\beta} - \eta(\mu(t), \mu(t - \Delta t)) \}}{\int_{\mathcal{M}} \exp \{ \mathbf{x}'(\mu) \boldsymbol{\beta} - \eta(\mu, \mu(t - \Delta t)) \} d\mu}$$



Movement model



Purposes:

- Account for autocorrelation
- Define availability distribution

$$\mu(t - \Delta t)$$

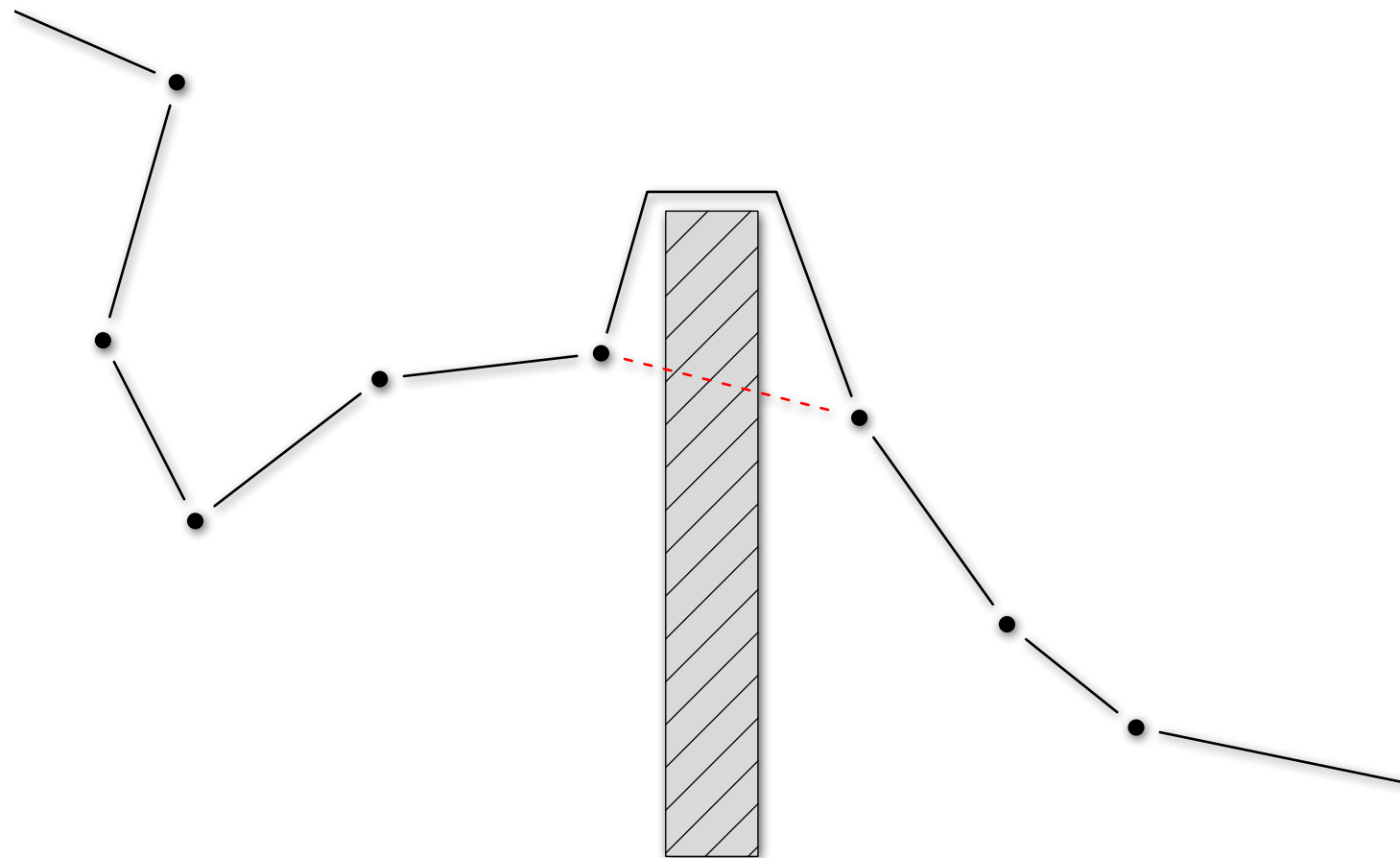
$$\mu(t - \Delta t)$$

Large Δt

Small Δt

Process model

$$\mu(t) \sim \frac{\exp \{ \mathbf{x}'(\mu(t)) \beta - \eta(\mu(t), \mu(t - \Delta t)) \}}{\int_{\mathcal{M}} \exp \{ \mathbf{x}'(\mu) \beta - \eta(\mu, \mu(t - \Delta t)) \} d\mu}$$



Process model

$$\tilde{\mu}(t) \sim \sum_{j=1}^J \pi_j \delta_{\tilde{\mu}_j}$$

$\tilde{\mu}(t)$ — location of haul-out site at time t

$\tilde{\mu}_j$ — location of potential haul-out site

$\delta_{\tilde{\mu}_j}$ — point mass at $\tilde{\mu}_j$

π_j — mixing proportion for component j